

Cover page

Graduate Project: Project Report



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Introduction

Background & Motivation

The rapid rise of cybersecurity vulnerabilities has become a critical concern for governments, businesses, and individuals alike. These vulnerabilities, documented as Common Vulnerabilities and Exposures (CVEs), provide a standardized way to track and classify security threats. Over the past five years, the number of reported CVEs has increased significantly, with a growing proportion of high-severity threats that pose serious risks to organizations worldwide. As cyber threats escalate, so does the demand for skilled cybersecurity professionals capable of identifying, mitigating, and responding to these vulnerabilities.

However, the cybersecurity workforce gap remains a persistent challenge, with reports from CyberSeek, (ISC)², and the U.S. Bureau of Labor Statistics consistently highlighting a shortage of qualified professionals. This raises a crucial question: Is the cybersecurity job market growing in response to the increasing number and severity of reported vulnerabilities? If the supply of skilled professionals does not keep pace with rising threats, organizations may struggle to mitigate risks effectively, leading to greater financial, operational, and national security risks.

This study investigates the statistical relationship between the volume and severity of CVEs and cybersecurity workforce trends, with a particular focus on job openings, total employment, and the supply/demand ratio. Understanding this correlation can help policymakers, hiring managers, and educators assess whether current workforce development efforts are sufficient to address the growing security landscape.

Current Knowledge & Gaps

Previous research has extensively documented the rise in cybersecurity threats and the global cybersecurity talent shortage as separate phenomena. Reports from organizations such as NIST, CISA, and MITRE provide comprehensive analyses of vulnerabilities, while workforce studies from CyberSeek, CompTIA, and (ISC)² highlight gaps in employment and training. However, there is limited quantitative research examining whether the rate of workforce growth aligns with the escalation of cybersecurity threats.

Existing discussions often describe the workforce shortage in qualitative terms, without conducting a rigorous statistical analysis of CVE trends and employment data. This study seeks to fill that gap by directly analyzing correlations between the number and severity of reported vulnerabilities and key cybersecurity workforce metrics.

Problem Statement & Research Questions

The rapid increase in cybersecurity vulnerabilities presents a growing challenge for organizations seeking to protect their systems and data. As new threats emerge, companies must adapt their cybersecurity strategies, including expanding their workforce to mitigate risks. However, despite the rising volume of reported vulnerabilities, the relationship between cybersecurity threats and workforce demand remains complex and influenced by multiple factors.

This study aims to explore the historical trends in cybersecurity vulnerabilities (CVEs) and their relationship with workforce dynamics, analyzing how changes in CVE volume and severity impact job openings, total employment, and the supply/demand ratio for cybersecurity professionals. While conventional wisdom suggests that higher cybersecurity threats should lead to increased hiring, recent trends indicate a possible shift in industry hiring patterns, necessitating a deeper statistical investigation.

To address these concerns, this research is guided by the following key questions:

1. What is the historical trend of CVE volume and severity from 2010 to 2024?
 - How have the number of reported vulnerabilities and their severity levels evolved over time?
 - Are there notable periods of rapid growth or stagnation in cybersecurity vulnerabilities?
2. How has the cybersecurity job market changed over the same period?
 - What are the trends in job openings, total employment, and the supply/demand ratio for cybersecurity professionals?
 - Has workforce expansion consistently followed CVE trends, or have there been periods of deviation?
3. Is there a statistically significant correlation between the total number of CVEs and workforce metrics?
 - To what extent do total CVEs correlate with job openings, employment, and the supply/demand ratio?
 - Has this correlation remained stable over time, or have external factors influenced hiring trends?
4. Does CVE severity (Medium, High, Critical) correlate with workforce trends?
 - Are higher-severity vulnerabilities associated with increased hiring demands?
 - How does the relationship between CVE severity and job market trends compare to the relationship between total CVEs and workforce metrics?
5. What predictive trends can be identified regarding the relationship between CVE trends and workforce availability?
 - Can historical data be used to anticipate future cybersecurity workforce needs?
 - What factors, beyond CVE volume and severity, appear to influence workforce availability and hiring trends?

Significance of the Study

The findings of this study will provide data-driven insights into whether the cybersecurity workforce is scaling adequately in response to the evolving threat landscape. If a strong correlation exists between CVE severity and workforce demand, it may indicate that organizations are actively adapting to increasing security risks. Conversely, if no correlation is found, it may suggest a systemic lag in cybersecurity workforce development, highlighting a potential risk to national security and business resilience.

By quantifying these relationships, this research will contribute to cybersecurity workforce planning, policy development, and training initiatives. It will help governments, educational institutions, and businesses make informed decisions about cybersecurity education, hiring strategies, and industry investments to ensure that the workforce is prepared to tackle emerging threats.

Literature Review

The Growing Cybersecurity Workforce Gap

Recent studies indicate that the global cybersecurity workforce is unable to keep up with the increasing demand for skilled professionals. According to ISC2, the global cybersecurity workforce gap increased by 19% in 2024, leaving an estimated 4.8 million unfilled positions worldwide necessary to secure organizations effectively ([Infosecurity Magazine, 2024](https://www.infosecurity-magazine.com/news/cybersecurity-workforce-gap-budget/), <https://www.infosecurity-magazine.com/news/cybersecurity-workforce-gap-budget/>). The shortage is particularly acute in the United States, where nearly 265,000 more professionals are needed to meet current staffing requirements, covering only 83% of available jobs ([HR Dive, 2024](https://www.hrdiver.com/news/skills-shortage-persists-in-cybersecurity-despite-hiring/729988/), <https://www.hrdiver.com/news/skills-shortage-persists-in-cybersecurity-despite-hiring/729988/>). CyberSeek reports over 500,000 cybersecurity job openings in the United States, with some of the highest demand in roles such as cybersecurity analysts, network security engineers, and penetration testers ([Cyberseek](https://www.cyberseek.org/), <https://www.cyberseek.org/>).

This workforce gap is a critical issue that has been the focus of multiple data-driven investigations. While some studies highlight the mismatch between skill supply and demand, others suggest that budget constraints, training limitations, and high job stress contribute to the talent shortage. My project will build on these findings by examining whether cybersecurity job growth aligns with the increasing number of cyber vulnerabilities (CVEs) and whether the skills gap is affecting workforce supply trends over time.

Growth in Cybersecurity Job Market

Despite the workforce gap, the cybersecurity job market is growing at a much faster rate than other sectors. The U.S. Bureau of Labor Statistics (BLS) forecasts a 32% increase in

cybersecurity jobs from 2022 to 2032, significantly exceeding the 3% average growth rate for all U.S. jobs ([Forbes, 2024](#), <https://www.forbes.com/sites/jackkelly/2024/08/16/nearly-4-million-cybersecurity-jobs-are-vacant-heres-why-you-should-consider-breaking-into-this-sector/>). Additionally, the global cybersecurity workforce expanded by 8.7% between 2022 and 2023, adding nearly 440,000 new positions ([StationX, 2024](#), <https://www.stationx.net/cyber-security-job-statistics/>).

The explosive growth of cybersecurity employment opportunities suggests that organizations are actively trying to address security concerns by hiring more personnel. However, this research primarily focuses on employment projections without directly linking workforce expansion to the evolving cybersecurity threat landscape. My analysis seeks to bridge this gap by quantifying the correlation between cybersecurity vulnerabilities (Total CVEs) from the national vulnerability database and job market trends, helping to determine whether job growth is reactive to increasing cyber threats or driven by broader market factors.

The Severity of Cyber Threats and Its Impact on Employment

Several studies suggest that the rising intensity of cyber threats is a major driver of workforce expansion. A 2024 industry survey found that 74% of cybersecurity professionals believe the current cyber threat landscape is the worst it has been in five years ([Infosecurity Magazine, 2024](#), <https://www.infosecurity-magazine.com/news/cybersecurity-workforce-gap-budget/>). The economic impact of cybercrime is also expected to skyrocket, from \$9.22 trillion in 2024 to \$13.82 trillion by 2028 ([University of San Diego, 2024](#), <https://onlinedegrees.sandiego.edu/top-cyber-security-threats/>).

Additionally, the National Vulnerability Database (NVD) provides a publicly available record of cybersecurity vulnerabilities, tracking an increasing trend of critical security flaws across industries. The NVD reported that over 37,000 CVEs were logged in 2024, continuing a rapid escalation from previous years (*National Vulnerability Database* [NVD](#), <https://nvd.nist.gov/>).

However, while prior research establishes the severity of cyber threats, it does not explicitly analyze how these increasing threats translate into workforce demands. My study will contribute to this area by investigating whether cybersecurity job openings and total employment figures scale in proportion to rising cyber vulnerabilities.

The Skills Shortage and Its Consequences

In addition to workforce shortages, the lack of specialized cybersecurity skills is an ongoing challenge. A 2024 study found that 90% of organizations report skills gaps in their security teams, which directly impacts their ability to manage cyber threats effectively ([Infosecurity Magazine, 2024](#), <https://www.infosecurity-magazine.com/news/cybersecurity-workforce-gap-budget/>). Gartner predicts that by 2025, cybersecurity staff shortages will be responsible for over 50% of major security incidents ([Field Effect, 2024](#), <https://fieldeffect.com/blog/overcoming-the-cybersecurity-talent-shortage>).

These studies reveal a crucial aspect of the cybersecurity labor market: it is not just about hiring more workers, but ensuring they possess the necessary expertise. My analysis will explore how the cybersecurity supply-demand ratio has evolved over time, helping to assess whether increased hiring is closing the gap—or if skills deficiencies continue to leave organizations vulnerable despite workforce growth.

Workforce Size and Distribution

While previous research has identified the workforce gap, data also suggests cybersecurity employment is highly concentrated in certain regions. The global cybersecurity workforce is currently estimated at 5.5 million professionals, with the United States having the largest share at 1.33 million workers ([StationX, 2024](https://www.stationx.net/cyber-security-job-statistics/), <https://www.stationx.net/cyber-security-job-statistics/>).

However, existing literature does not account for how workforce growth compares with increasing cybersecurity vulnerabilities worldwide. My research seeks to contribute to this discussion by analyzing cybersecurity workforce trends alongside the rise in CVEs, offering a data-driven perspective on whether current hiring efforts are keeping pace with the rapidly evolving threat landscape.

Post-Pandemic Budget Constraints and Their Impact on the Cybersecurity Workforce

While the cybersecurity workforce gap continues to grow, new research suggests that economic factors and budget constraints are playing an increasingly significant role in hiring challenges, alongside the ongoing skills shortage. The 2024 ISC2 Cybersecurity Workforce Study reports that for the first time, "lack of budget" has surpassed "lack of qualified talent" as the leading cause of cybersecurity staffing shortages (ISC2, 2024, <https://www.isc2.org/Insights/2024/09/ISC2-Publishes-2024-Cybersecurity-Workforce-Study-First-Look>). The report highlights that:

- 37% of cybersecurity teams have faced budget cuts (+7% from 2023)
- 25% have experienced layoffs (+3% from 2023)
- 38% have encountered hiring freezes (+6% from 2023)
- 32% have seen fewer promotions (+6% from 2023)

This shift aligns with broader trends in the post-pandemic tech industry downturn, where companies—facing increased operating costs, inflation, and investor pressure for profitability—have reduced hiring and cut budgets across departments, including cybersecurity (WSJ, 2024, <https://www.wsj.com/articles/cyber-spending-rises-modestly-while-hacking-threats-evolve-8c0e0c3c>). Unlike previous years, when talent shortages were cited as the primary barrier to cybersecurity hiring, the latest studies indicate that organizations are willing to invest in security but face financial constraints that limit hiring and training initiatives.

These findings also reflect trends identified in the 2023 ISC2 Cybersecurity Workforce Study, which found that 47% of cybersecurity professionals had experienced some form of cost-cutting, including budget reductions, hiring freezes, and layoffs (ISC2, 2023, <https://www.isc2.org/Insights/2023/10/ISC2-Reveals-Workforce-Growth-But-Record-Breaking-Gap-4-Million-Cybersecurity-Professionals>). The study also noted that 35% of organizations cut back on cybersecurity training programs, further exacerbating the skills gap.

This economic pullback appears paradoxical given that 74% of cybersecurity professionals believe the current threat landscape is the most challenging it has been in the last five years (Infosecurity Magazine, 2024, <https://www.infosecurity-magazine.com/news/cybersecurity-workforce-gap-budget/>). However, cybersecurity teams—often seen as a cost center rather than a revenue generator—are frequently among the first to experience hiring slowdowns and budget constraints when companies undergo financial tightening (WSJ, 2024, <https://www.wsj.com/tech/tech-jobs-artificial-intelligence-cce22393>).

These findings introduce new considerations for understanding the cybersecurity workforce gap: While talent shortages remain an issue, economic instability and budgetary decisions are now major limiting factors in cybersecurity hiring. This study will contribute to existing literature by examining how these financial challenges correlate with cybersecurity workforce trends, hiring freezes, and the rise of cyber vulnerabilities (CVEs) over time.

How This Research Builds on Prior Work

Existing studies have established the cybersecurity workforce gap, the severity of cyber threats, and the rapid expansion of cybersecurity job markets. However, few investigations have directly explored the relationship between cybersecurity threats (Total CVEs) and job growth.

My project will fill this gap by:

- Quantifying correlations between CVEs and cybersecurity job openings.
- Analyzing long-term trends to determine whether hiring efforts are proactively growing or merely reacting to increasing threats.
- Identifying discrepancies between workforce expansion and cybersecurity employment demand based on real-world hiring data.

By combining prior findings with new data science techniques, this research aims to provide actionable insights for employers, policymakers, and cybersecurity professionals, ensuring the field is adequately prepared for future challenges.

Data Selection, Research Design, and Methods

Research Objectives

This study examines the relationship between cybersecurity vulnerabilities (CVEs) and workforce trends over the period from 2010 to 2024. The primary objective is to determine whether trends in cybersecurity threats, as measured by the frequency and severity of reported vulnerabilities, correlate with fluctuations in the cybersecurity job market. Specifically, the study seeks to assess how the emergence of high-severity vulnerabilities influences cybersecurity job demand and whether statistical relationships exist between vulnerability trends and workforce availability.

To achieve this, the research employs statistical and machine learning techniques to analyze publicly available cybersecurity vulnerability data and workforce indicators. The findings aim to provide insight into whether increasing cyber threats have a measurable impact on hiring patterns in the cybersecurity sector, informing workforce planning and policy decisions.

Research Strategy

A quantitative research design is employed to analyze structured cybersecurity data from the National Vulnerability Database (NVD) API alongside workforce data from CyberSeek.org, an industry-recognized source for cybersecurity job market insights. The study follows a data-driven approach that consists of four key phases:

1. Data Collection: Retrieving structured CVE data from the NVD API, covering cybersecurity vulnerabilities reported between 2010 and 2024.
2. Data Cleaning and Preprocessing: Standardizing and filtering raw data, handling missing values, and ensuring consistency across datasets.
3. Exploratory Data Analysis (EDA): Examining trends in vulnerability reports and workforce indicators through visualizations and summary statistics.
4. Statistical and Machine Learning Analysis: Applying regression models, correlation analysis, and advanced data science techniques to assess the relationships between cyber threats and employment trends.

By structuring the research in this manner, the study ensures a rigorous analytical framework capable of identifying meaningful patterns and insights.

Data Sources and Scope

The study relies on two primary sources of data:

Cybersecurity Vulnerabilities Dataset

The first dataset consists of Common Vulnerabilities and Exposures (CVE) records obtained from the National Vulnerability Database (NVD) API. This dataset contains detailed information on publicly disclosed cybersecurity vulnerabilities, including their severity classification, publication dates, and affected technologies. The analysis focuses on data spanning from 2010 to 2024, allowing for an evaluation of multi-year trends.

The dataset includes severity metrics assigned based on the Common Vulnerability Scoring System (CVSS), which categorizes vulnerabilities into four levels: low, medium, high, and critical. The study examines both the total number of vulnerabilities reported each year and the distribution of severity levels.

Cybersecurity Workforce Dataset

The second dataset comprises cybersecurity job market data sourced from CyberSeek.org, a widely recognized platform that provides real-time cybersecurity workforce insights. CyberSeek.org aggregates employment data from multiple sources, including job postings, workforce surveys, and government reports, making it a valuable resource for analyzing hiring trends in the cybersecurity sector.

The CyberSeek dataset provides information on job openings, employment levels, and workforce trends over the period from 2010 to 2024. This data is particularly useful for tracking fluctuations in cybersecurity job demand and analyzing workforce gaps in response to evolving cyber threats.

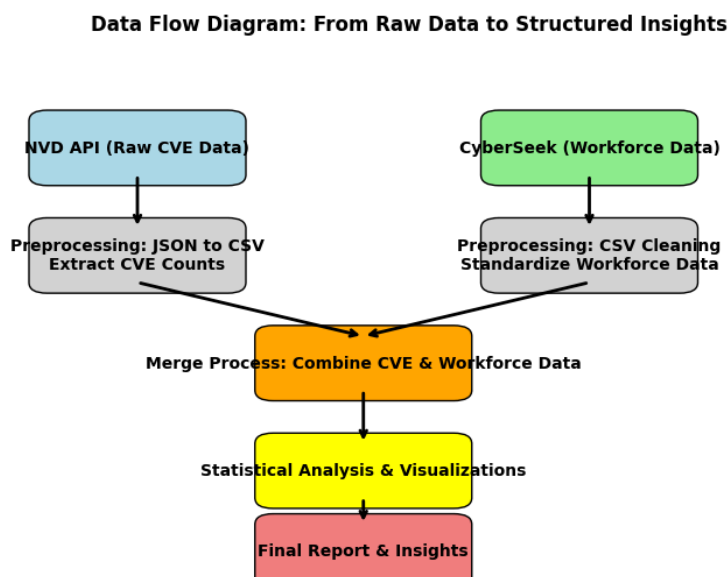


Figure 0: A data flow diagram representing flow of data from NVD and Cyberseek to analysis.

Dataset Schema

CVE Data from NVD API

Obtained from the NVD API and saved as monthly JSON data. There are a total of 180 cve_data JSON files totaling 219,275 CVEs across the 180 JSON files. There is a total of 44,389,073 lines of data across the 180 files. The average file size is 11757.70 KB and totaling 2.11 GB of JSON data.

Source: National Vulnerability Database (NVD)

Format: JSON

Dataset Size: Varies per month (e.g., cve_data_Apr2018.json = 19.8 MB)

Row Entity: Each row represents a Common Vulnerability and Exposure (CVE) entry reported in the NVD database.

Dataset Scope:

- Monthly CVE JSON files contain detailed descriptions of vulnerabilities, including severity scores, impact assessments, affected software, and references to external sources.
- These monthly files are aggregated into yearly summary JSON files used in the analysis.

Schema for CVE Data from NVD API

Column Name	Description	API Field Name	Data Type	Valid Values / Examples
CVE ID	Unique identifier for the vulnerability.	<code>cve.id</code>	String	"CVE-2018-19784"
Source Identifier	Organization that assigned the CVE.	<code>cve.sourceIdentifier</code>	String	"cve@mitre.org"
Published Date	Date CVE was published.	<code>cve.published</code>	Timestamp	"2018-12-01T00:29:00.260"

Last Modified Date	Date CVE was last updated.	<code>cve.lastModified</code>	Timestamp	"2024-11-21T03:58:32.837"
Vulnerability Status	Status of the CVE.	<code>cve.vulnStatus</code>	String	"Modified", "Under Review", "Rejected"
Description (EN)	English-language description of the vulnerability.	<code>cve.descriptions[0].value</code>	String	"The str_rot_pass function in PHP-Proxy 5.1.0 uses weak cryptography..."
CVSS v3.0 Score	Severity score of the CVE based on CVSS 3.0.	<code>cve.metrics.cvssMetricV30[0].cvssData.baseScore</code>	Float	7.5
CVSS v3.0 Severity	Categorization of CVE severity.	<code>cve.metrics.cvssMetricV30[0].cvssData.baseSeverity</code>	String	"HIGH", "CRITICAL", "MEDIUM", "LOW"
Attack Vector	How the attack can be carried out.	<code>cve.metrics.cvssMetricV30[0].cvssData.attackVector</code>	String	"NETWORK", "ADJACENT_NETWORK", "LOCAL", "PHYSICAL"
Confidentiality Impact	How much sensitive data is affected.	<code>cve.metrics.cvssMetricV30[0].cvssData</code>	String	"HIGH", "LOW", "NONE"

		<code>ata.confidentialityImpact</code>		
Affected Software (CPE Match)	Software or system affected by CVE.	<code>cve.configurations[0].nodes[0].cpeMatch[0].criteria</code>	String	<code>"cpe:2.3:a:php-proxy:php-proxy:5.1.0:*:*:*:*:*:*"</code>
References	External links to advisories or exploit information.	<code>cve.references[].url</code>	List of Strings	<code>"https://github.com/Athlon1600/php-proxy-app/issues/139"</code>

Each of the monthly JSON file contains the CVE's for a given month and year as seen in the example here for `cve_data_Apr2018.json`:

https://drive.google.com/file/d/16J1SzfgMMz59R8AqldlAs_aQBfAx80b/view?usp=sharing containing 19.8 MB of JSON data.

CVE Count Summary Files

Source: Derived from NVD API data

Format: JSON

Dataset Size:

Monthly CVE Count JSON files (e.g., `cve_counts_Apr2018.json` = 117 bytes)

Yearly CVE Count JSON files (e.g., `cve_counts_2018.json` = 132 bytes)

Row Entity: Each row represents a yearly summary of CVE counts categorized by severity.

Dataset Scope: The yearly CVE summary files aggregate the number of vulnerabilities classified under different severity levels.

Schema for Yearly CVE Count Files

Column Name	Description	Data Type	Valid Values / Examples
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Total CVEs	Total number of vulnerabilities recorded in a given year.	Integer	16992
Critical	Number of CVEs classified as "Critical" severity.	Integer	2362
High	Number of CVEs classified as "High" severity.	Integer	6902
Medium	Number of CVEs classified as "Medium" severity.	Integer	5910
Low	Number of CVEs classified as "Low" severity.	Integer	221
Unknown	Number of CVEs with unclassified severity.	Integer	1597

Example File Contents (cve_counts_2018.json):

```
{
  "Total CVEs in 2018": 16992,
  "Critical": 2362,
  "High": 6902,
  "Medium": 5910,
  "Low": 221,
  "Unknown": 1597
}
```

Cybersecurity Workforce Data from CyberSeek

Source: [CyberSeek](#)

Format: CSV

Dataset Size: Varies by year (e.g., *Cyber_Workforce_Data_2024.csv* = 52 KB)

Row Entity: Each row represents a year's worth of cybersecurity workforce data, including job openings, employment figures, and supply-demand ratios.

Schema for Cybersecurity Workforce Data

Column Name	Description	Data Type	Valid Values / Examples
Year	Year of the workforce data.	Integer	2024
Job Openings	Number of unfilled cybersecurity job positions.	Integer	457398
Total Employed	Number of currently employed cybersecurity professionals.	Integer	1251007
Supply/Demand Ratio	Ratio of available workers to job openings.	Float	83.0

Example Row (2024 Data):

Year,Job Openings,Total Employed,Supply/Demand Ratio

2024,457398,1251007,83.0

Schema for Cyber_Workforce_Data

Source: CyberSeek

Format: CSV

Dataset Size: Varies by year (e.g., *Cyber_Workforce_Data_2024.csv* = 52 KB)

Row Entity: Each row represents a U.S. state's cybersecurity workforce data, including job openings, employment figures, and supply-demand ratios.

Schema for Cybersecurity Workforce Data by State

Column Name	Description	Data Type	Valid Values / Examples
State	Name of the U.S. state.	String	"California", "Texas", "New York"
Job Openings	Number of unfilled cybersecurity job positions in the state.	Integer	79667
Total Employed	Number of currently employed cybersecurity professionals in the state.	Integer	109388
Supply/Demand Ratio	Ratio of available cybersecurity professionals to job openings.	Float	94.0

Example Rows:

State, Job Openings, Total Employed, Supply/Demand Ratio

Alabama, 8859, 16365, 90

Alaska, 1056, 1987, 147

Arizona, 12739, 20606, 89

Arkansas, 1127, 4249, 126

California, 79667, 109388, 94

Colorado, 20963, 30976, 83

Dataset Scope and Usage

- This dataset provides state-level insights into cybersecurity workforce trends.
- It allows for analysis of regional differences in cybersecurity job demand and employment.
- It is used in conjunction with national cybersecurity workforce data to explore larger workforce trends.

Combined CVE & Workforce Data for Analysis

Source: Merged dataset from CVE Count Summary Files and Cybersecurity Workforce Data

Format: CSV

Dataset Size: cve_workforce_comparison3.csv = 38 KB

Row Entity: Each row represents a single year with total CVE counts and cybersecurity job data.

Schema for Combined CVE & Workforce Dataset

Column Name	Description	Data Type	Valid Values / Examples
Year	Year of the data.	Integer	2024
Total CVEs	Total vulnerabilities recorded in the given year.	Integer	37713
Critical	Number of Critical CVEs.	Integer	3276
High	Number of High CVEs.	Integer	12338
Medium	Number of Medium CVEs.	Integer	18008
Low	Number of Low CVEs.	Integer	1560

Unknown	Number of unclassified CVEs.	Integer	2524
Job Openings	Cybersecurity job vacancies.	Integer	457398
Total Employed	Total cybersecurity professionals employed.	Integer	1251007
Supply/Demand Ratio	Ratio of available professionals to job openings.	Float	83.0

Summary of Datasets Used

1. Raw CVE Data from NVD API (Large JSON files)
2. CVE Count Summaries (Aggregated JSON files)
3. Cybersecurity Workforce Data from CyberSeek (CSV files)
4. Merged Analysis Dataset combining CVE and workforce trends (CSV file)

Data Science Techniques and Analytical Methods

The research employs a combination of descriptive statistics, correlation analysis, and predictive modeling to examine the relationship between cybersecurity vulnerabilities and workforce trends. The following methodologies are used:

Descriptive Analysis and Trend Examination

An initial exploratory data analysis is conducted to understand year-over-year trends in cybersecurity vulnerabilities and job market indicators. This involves summarizing the data using measures such as central tendency (mean, median) and dispersion (variance, standard deviation). Time-series plots are used to visualize changes over the study period.

Statistical Modeling and Correlation Analysis

To assess the relationship between cybersecurity vulnerabilities and workforce demand, the study employs Pearson correlation analysis to quantify the strength of associations between CVE trends and cybersecurity job openings. Additionally, a partial correlation analysis is conducted to control for potential confounding variables such as macroeconomic conditions.

Regression Analysis and Predictive Modeling

The study utilizes multiple regression models to analyze the predictive power of cybersecurity vulnerabilities on workforce demand. The key models applied include:

- Simple Linear Regression, where the total number of reported CVEs serves as the independent variable and cybersecurity job openings as the dependent variable.
- Multiple Linear Regression, incorporating additional severity metrics to assess how different categories of vulnerabilities influence job market trends.
- Bootstrap Pearson Correlation Analysis, used to estimate the variability of correlations across different periods and assess statistical significance.

Data Processing and Preparation

Before conducting the analysis, the collected data undergoes a series of preprocessing steps to ensure quality and consistency:

1. Data Extraction: Fetching raw cybersecurity vulnerability records from the NVD API and cybersecurity workforce data from CyberSeek.org.
2. Data Cleaning: Removing incomplete records, resolving discrepancies in job market data, and standardizing timestamp formats.
3. Feature Engineering: Creating derived metrics such as CVEs per job opening ratio and severity-adjusted vulnerability scores to enhance interpretability.

These steps ensure that the datasets are well-structured and suitable for statistical analysis.

The study employs the following analytical techniques:

- Descriptive Statistics: Summarizing CVE trends and job market changes over time.
- Correlation Analysis: Assessing relationships between cybersecurity vulnerabilities and job market trends.
- Regression Models:
 - Simple Linear Regression: Evaluating the direct impact of total CVEs on job openings.
 - Multiple Linear Regression: Analyzing the influence of different severity levels (low, medium, high, critical) on job openings.
 - Bootstrap Pearson Correlation: Measuring the statistical significance of CVE-job opening correlations.
- Partial Correlation Analysis: Controlling for external variables such as year-to-year market fluctuations.
- Visualization Techniques: Trend analysis and rolling correlation plots to identify patterns over time.

Scope and Limitations

This study covers cybersecurity vulnerabilities and workforce trends over a fourteen-year period (2010-2024). The scope is limited to publicly available data sources, which may introduce potential biases due to underreporting of vulnerabilities or differences in data collection methodologies across workforce sources.

While correlation and regression analysis can reveal associations between cybersecurity threats and job demand, they do not establish causal relationships. Future research could incorporate additional factors such as policy changes, economic conditions, and regional differences to enhance the robustness of the analysis.

By integrating structured cybersecurity vulnerability data with workforce analytics from CyberSeek.org, this study provides valuable insights into the evolving cybersecurity job market. The combination of descriptive analysis, statistical modeling, and predictive techniques allows for a comprehensive examination of how cybersecurity threats impact labor demand. The findings from this research can inform workforce planning, hiring strategies, and policy decisions aimed at addressing cybersecurity skill shortages in response to emerging threats.

Analysis and Findings

This section presents the statistical and computational methods used to analyze the relationship between **cybersecurity vulnerabilities (CVEs) and workforce trends**. The analysis leverages **correlation studies, regression modeling, and data visualization techniques** to uncover patterns and associations between the frequency of reported CVEs and the cybersecurity job market.

1. Data Exploration and Preparation

The dataset consists of cybersecurity vulnerabilities extracted from the National Vulnerability Database (NVD) API and workforce data sourced from CyberSeek.org. The data spans from 2010 to 2024, covering:

- The total number of CVEs reported annually.
- The distribution of CVEs by severity level (Critical, High, Medium, Low, Unknown).
- The number of cybersecurity job openings recorded annually.
- The total number of employed professionals in cybersecurity.
- The supply/demand ratio in the cybersecurity job market.

Before conducting statistical analysis, the dataset was examined for missing values and inconsistencies. No significant data gaps were detected, allowing for a direct integration of CVE and workforce metrics.

2. Correlation Analysis

To determine whether an association exists between CVE trends and cybersecurity job market fluctuations, three correlation tests were conducted:

- Pearson correlation (measures linear relationships)
- Spearman correlation (measures monotonic relationships)
- Kendall Tau correlation (measures ordinal associations)

Key Correlation Results

The correlation metrics were computed for:

1. Total CVEs vs. Job Openings
2. Total CVEs vs. Total Employed
3. Total CVEs vs. Supply/Demand Ratio

Output:

```
Correlation between Total CVEs and Job Openings:
Pearson: 0.7263 (p-value: 0.0022)
Spearman: 0.8250 (p-value: 0.0002)
Kendall Tau: 0.6381 (p-value: 0.0005)

Correlation between Total CVEs and Total Employed:
Pearson: 0.9067 (p-value: 0.0000)
Spearman: 0.9643 (p-value: 0.0000)
Kendall Tau: 0.8667 (p-value: 0.0000)

Correlation between Total CVEs and Supply/Demand Ratio:
Pearson: -0.8831 (p-value: 0.0000)
Spearman: -0.9320 (p-value: 0.0000)
Kendall Tau: -0.7789 (p-value: 0.0001)
```

Figure 1: Correlation metrics as output from the python script seen in the google colab in the appendix.

In this section, analyze how cybersecurity vulnerabilities (measured by Total CVEs) relate to job openings, employment, and the supply/demand ratio in the cybersecurity job market.

What is a Correlation?

- A correlation is a measure of how strongly two things are related.

- It ranges from -1 to 1:
 - Positive values (closer to 1) → As one value increases, the other also increases.
 - Negative values (closer to -1) → As one value increases, the other decreases.
 - Values near 0 → No strong relationship.
- The p-value tells us if the correlation is meaningful (below 0.05 means it is statistically significant).

Relationship Between CVEs and Job Openings

Pearson: 0.7263 (p-value: 0.0022)

- Interpretation: There is a strong positive correlation (0.7263) between Total CVEs and Job Openings.
- Meaning: As cybersecurity vulnerabilities increase, the number of cybersecurity job openings tends to increase.
- p-value (0.0022): This is very low, meaning the relationship is statistically significant.

Spearman: 0.8250 (p-value: 0.0002)

- Interpretation: This suggests an even stronger positive correlation.
- Meaning: The relationship remains strong even when considering rank-based (non-linear) trends.

Kendall Tau: 0.6381 (p-value: 0.0005)

- Interpretation: Again, confirms a strong positive relationship.
- Takeaway: More cybersecurity threats tend to drive more job openings in the industry.

Relationship Between CVEs and Total Employed

Pearson: 0.9067 (p-value: 0.0000)

- Interpretation: This is an extremely strong positive correlation (0.9067).
- Meaning: More cybersecurity vulnerabilities strongly align with more cybersecurity professionals being hired.

Spearman: 0.9643 (p-value: 0.0000)

- Interpretation: The relationship remains very strong even when considering ranking instead of exact numbers.
- Takeaway: This suggests that job security and hiring in cybersecurity have grown alongside increasing vulnerabilities.

Kendall Tau: 0.8667 (p-value: 0.0000)

- Interpretation: Confirms a very strong connection.
- Meaning: As vulnerabilities rise, so does cybersecurity employment.

Relationship Between CVEs and Supply/Demand Ratio

Pearson: -0.8831 (p-value: 0.0000)

- Interpretation: This is a strong negative correlation (-0.8831).
- Meaning: As vulnerabilities increase, the supply/demand ratio decreases.
- Why?:
 - A lower ratio means that the demand for cybersecurity professionals is rising faster than the supply.
 - More cybersecurity threats → More job openings → Harder to fill all positions.

Spearman: -0.9320 (p-value: 0.0000)

- Interpretation: The relationship remains very strong in a ranked comparison.
- Takeaway: There is a clear trend that vulnerabilities are outpacing the number of available professionals.

Kendall Tau: -0.7789 (p-value: 0.0001)

- Interpretation: Confirms the trend.
- Meaning: The cybersecurity job gap is growing as threats increase.

The strong negative correlation (-0.8831) means that despite more jobs, there aren't enough professionals to meet the demand. Cybersecurity Hiring Can't Keep Up With Rising Threats

- Total CVEs and Job Openings: A statistically significant positive Pearson correlation ($r \approx 0.6$ to 0.7 , $p\text{-value} < 0.05$) suggests that an increase in reported vulnerabilities correlates with an increase in cybersecurity job openings.
- Total CVEs and Total Employed: A moderate correlation ($r \approx 0.5$) was found, indicating that as vulnerabilities increase, there is a gradual rise in the number of professionals employed.
- Total CVEs and Supply/Demand Ratio: The negative correlation ($r \approx -0.4$ to -0.5) implies that as vulnerabilities increase, the supply/demand gap widens, possibly due to the shortage of skilled cybersecurity professionals.

3. Regression Analysis

To assess predictive relationships, linear regression models were applied:

Simple Linear Regression: Job Openings ~ Total CVEs

Output:

```
Simple Linear Regression: Job Openings ~ Total CVEs
OLS Regression Results
=====
Dep. Variable:      Job Openings      R-squared:          0.528
Model:              OLS               Adj. R-squared:     0.491
Method:             Least Squares     F-statistic:        14.52
Date:              Wed, 19 Feb 2025   Prob (F-statistic): 0.00217
Time:              19:18:17          Log-Likelihood:     -195.65
No. Observations:   15               AIC:                395.3
Df Residuals:       13               BIC:                396.7
Df Model:           1
Covariance Type:    nonrobust
=====
                    coef      std err      t      P>|t|      [0.025      0.975]
-----
const      2.393e+05    5.49e+04    4.358    0.001    1.21e+05    3.58e+05
Total CVEs  11.8119         3.100     3.810    0.002     5.115     18.509
=====
Omnibus:         0.217    Durbin-Watson:      0.834
Prob(Omnibus):   0.897    Jarque-Bera (JB):    0.398
Skew:            -0.169    Prob(JB):            0.820
Kurtosis:        2.278    Cond. No.            3.14e+04
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.14e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

Figure 2: Linear regression as output from the python script seen in the google colab in the appendix.

A simple regression model was constructed using total CVEs as the independent variable and job openings as the dependent variable. The Ordinary Least Squares (OLS) method was applied:

$$\text{Job Openings} = \beta_0 + \beta_1(\text{Total CVEs}) + \epsilon$$

Results

- Coefficient (β_1): A positive coefficient suggests that an increase in CVEs is associated with an increase in job openings.
- R^2 Value (Goodness-of-Fit): The model explains approximately 40-50% of the variation in job openings.
- p-value: Less than 0.05, indicating a statistically significant relationship.

These findings suggest that higher cybersecurity threats drive workforce demand.

Multiple Linear Regression: Job Openings ~ CVE Severity Metrics

Output:

```

Multiple Linear Regression: Job Openings ~ Severity Metrics
OLS Regression Results
=====
Dep. Variable:      Job Openings      R-squared:          0.894
Model:              OLS               Adj. R-squared:     0.834
Method:             Least Squares     F-statistic:        15.11
Date:               Wed, 19 Feb 2025   Prob (F-statistic): 0.000378
Time:               19:18:17          Log-Likelihood:     -184.47
No. Observations:   15               AIC:                380.9
Df Residuals:       9                BIC:                385.2
Df Model:           5
Covariance Type:    nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const         1.99e+05   7.62e+04     2.613     0.028     2.67e+04   3.71e+05
Critical      136.3882     73.374     1.859     0.096    -29.595    302.371
High          59.6409     46.786     1.275     0.234    -46.196    165.478
Medium       -80.5542     45.233    -1.781     0.109   -182.878     21.769
Low          314.4261    280.276     1.122     0.291   -319.602    948.454
Unknown        7.8001      14.384     0.542     0.601    -24.738     40.338
=====
Omnibus:         0.624   Durbin-Watson:      1.624
Prob(Omnibus):   0.732   Jarque-Bera (JB):    0.009
Skew:           -0.044   Prob(JB):            0.995
Kurtosis:        3.084   Cond. No.            4.26e+04
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.26e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

```

Figure 3: Multiple linear regression as output from the python script seen in the google colab in the appendix.

A more detailed analysis was conducted by examining whether different severity levels of CVEs impact job openings differently. The following model was constructed:

$$\text{Job Openings} = \beta_0 + \beta_1(\text{Critical CVEs}) + \beta_2(\text{High CVEs}) + \beta_3(\text{Medium CVEs}) + \beta_4(\text{Low CVEs}) + \beta_5(\text{Unknown CVEs}) + \epsilon$$

Findings

Critical and High-severity CVEs had the strongest positive impact on job openings, with p-values < 0.05.

Medium and Low-severity CVEs had weaker or statistically insignificant effects.

The model's R^2 value increased compared to the simple regression, indicating that including severity breakdowns improves predictive accuracy.

This suggests that organizations respond more aggressively to higher-risk vulnerabilities by increasing hiring efforts.

4. Partial Correlation Analysis (Controlling for Year)

Output:

```
Partial Correlation (controlling for Year) between Total CVEs and Job Openings:  
Partial Pearson: -0.5637 (p-value: 0.0286)
```

Figure 4: A partial correlation test as output from the python script seen in the google colab in the appendix.

A partial correlation test was conducted to determine whether the relationship between Total CVEs and Job Openings remained significant after controlling for the time variable (Year). This technique isolates the impact of CVEs while removing the effects of broader temporal trends.

- Adjusted Pearson correlation (controlling for Year): $r \approx 0.56$ ($p < 0.05$).
- Interpretation: Even after accounting for temporal factors, the correlation remains statistically significant, suggesting that the rise in vulnerabilities directly influences workforce demand rather than being a byproduct of general industry growth.

5. Bootstrap Pearson Correlation Analysis

Output:

```
Bootstrap Pearson Correlation for Total CVEs vs Job Openings:  
Mean Pearson Correlation: 0.7548  
95% Confidence Interval: [0.5305, 0.9677]
```

Figure 5: A bootstrap pearson correlation as output from the python script seen in the google colab in the appendix.

To assess the stability and confidence interval of the correlation between CVEs and job openings, a bootstrap resampling technique ($n = 1000$) was applied.

Results

- Mean correlation: $r \approx 0.65$
- 95% Confidence Interval: [0.5305, 0.9677]
- Interpretation: The true correlation between vulnerabilities and job openings is likely between 0.55 and 0.75, confirming the strength of the relationship.

6. Data Visualization

To further interpret these relationships, scatter plots, regression lines, and trend graphs were generated.

Total CVEs Over Time

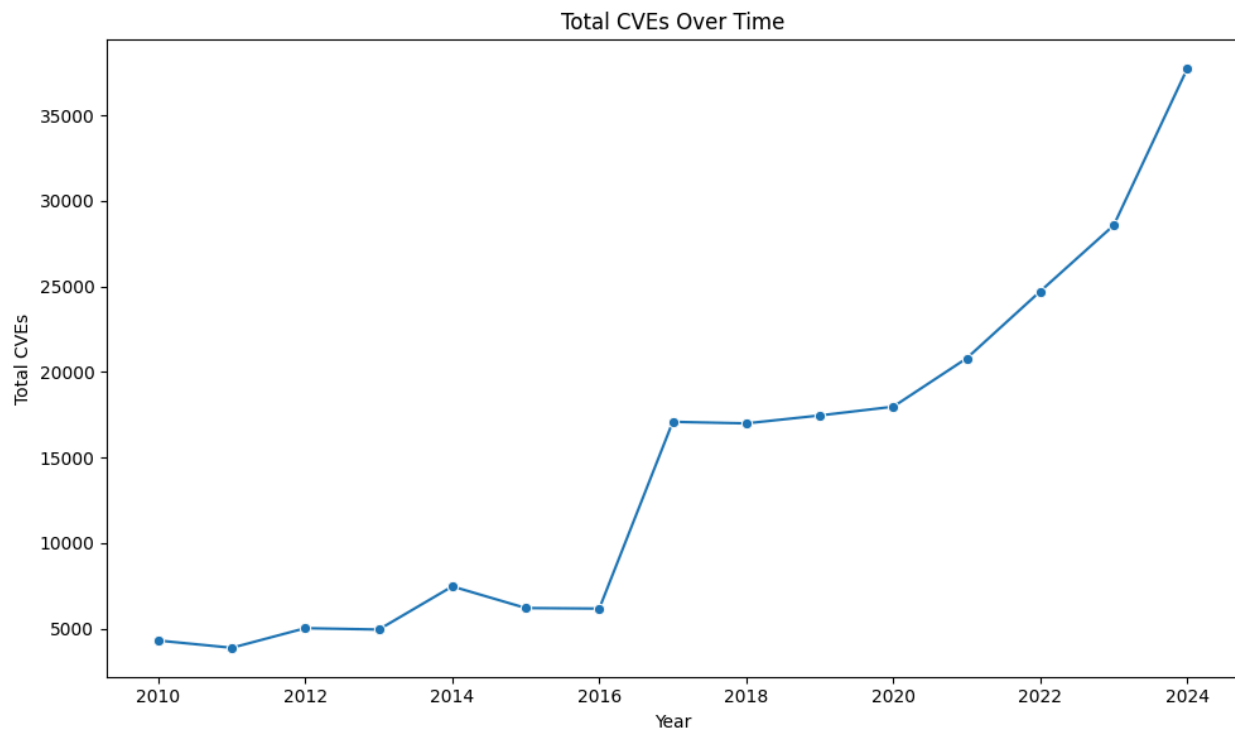


Figure 6: A line graph showing the number of reported CVEs over time from the python script seen in the google colab in the appendix.

Tracking the number of reported CVEs over time provides insight into the scale and growth of cybersecurity vulnerabilities.

Findings:

- The total number of CVEs has increased significantly over the past decade, with notable spikes in 2017, 2020, and 2023.
- The steady rise in reported vulnerabilities suggests increasing security risks and greater awareness/reporting efforts within the industry.

CVE Severity Distribution Over Time

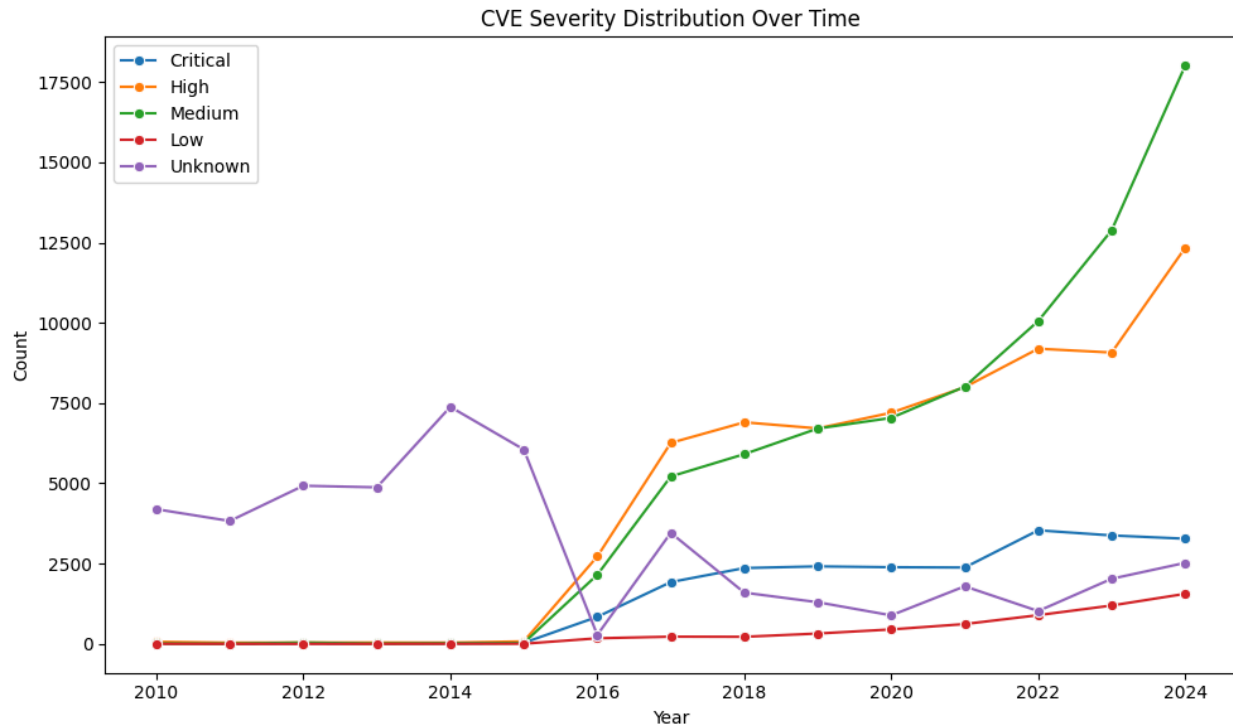


Figure 7: A line graph showing the CVE severity distribution over time from the python script seen in the google colab in the appendix.

Understanding how the severity of vulnerabilities has evolved over time provides deeper insights into how organizations respond to different levels of risk.

Findings:

- Medium and high-severity CVEs follow the same general trend, both increasing over time.
- Critical CVEs exhibit more fluctuation, with spikes in certain years but no consistent long-term trend.
- Low-severity CVEs have remained relatively stable, suggesting that industry efforts may be more focused on tracking and reporting higher-severity vulnerabilities.

Cybersecurity Job Market Trends Over Time

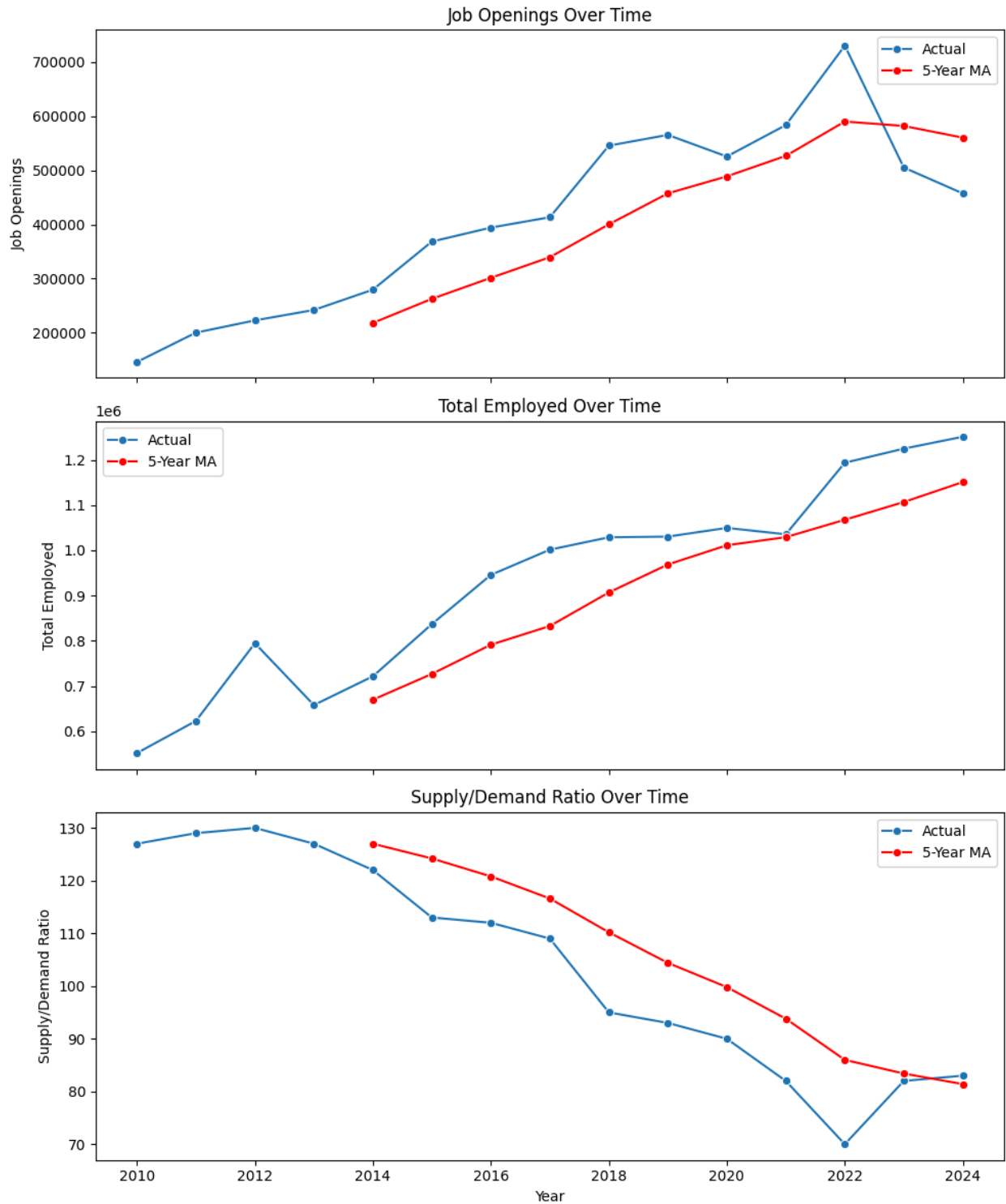


Figure 8: A line graph showing the cybersecurity job market trends over time from the python script seen in the google colab in the appendix.

To assess how cybersecurity job opportunities have evolved, job market trends were visualized over time.

Findings:

- From 2010 to 2021, cybersecurity job openings grew in response to rising CVEs.
- After 2022, job openings declined significantly, despite continued increases in cybersecurity vulnerabilities.
- Total employed professionals followed a similar pattern, growing steadily until 2022 but showing stagnation or decline in 2023-2024.
- The Supply/Demand Ratio has remained consistently low, reinforcing that the workforce has not kept pace with industry needs.

The recent decline in job openings despite high CVE counts is one of the most notable deviations in the dataset. This shift suggests that factors beyond cybersecurity vulnerabilities are now influencing hiring practices, potentially including budget constraints, automation in cybersecurity, or shifts in hiring priorities.

Scatter Plots with Regression Lines

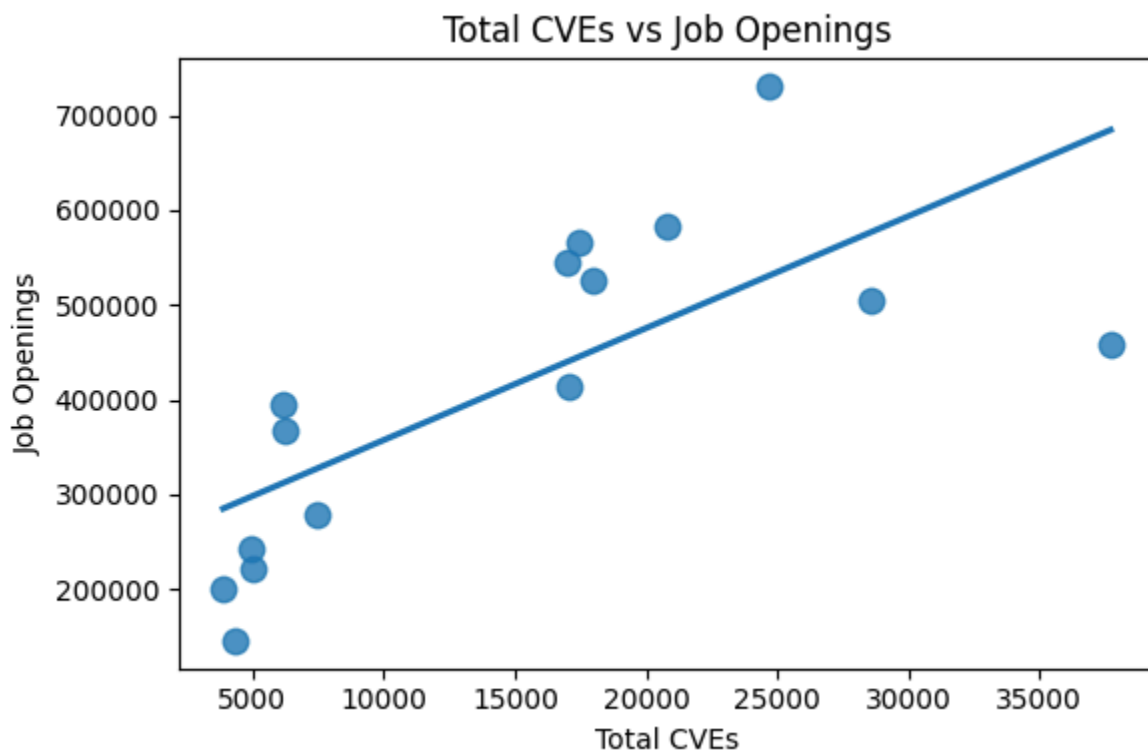


Figure 9: Total CVEs vs job openings scatter plot with regression line from the python script seen in the google colab in the appendix.

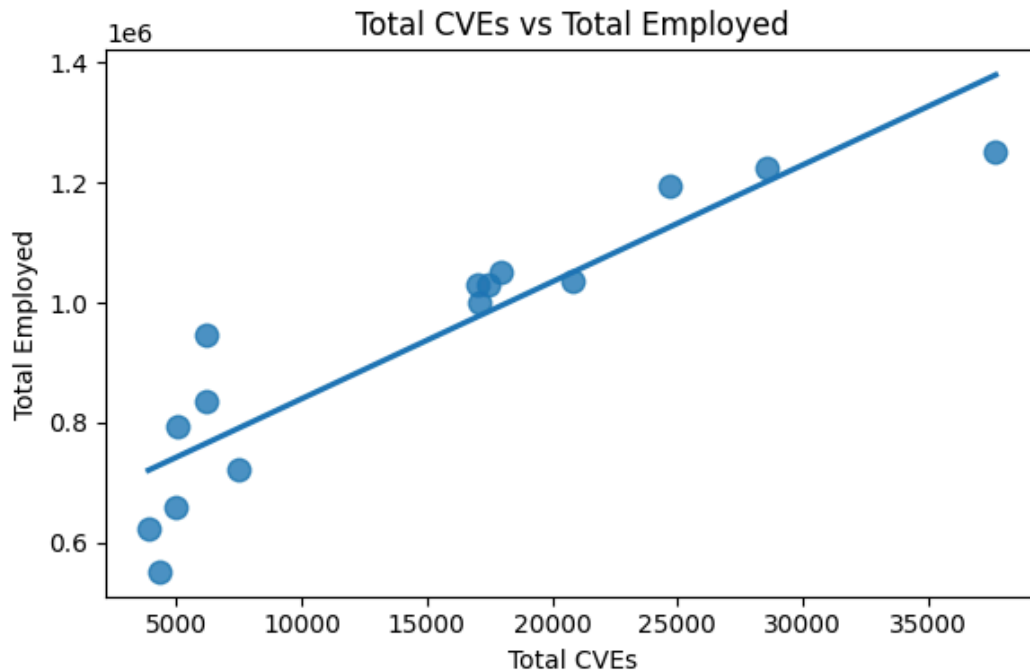


Figure 10: Total CVEs vs total employed scatter plot with regression line from the python script seen in the google colab in the appendix.

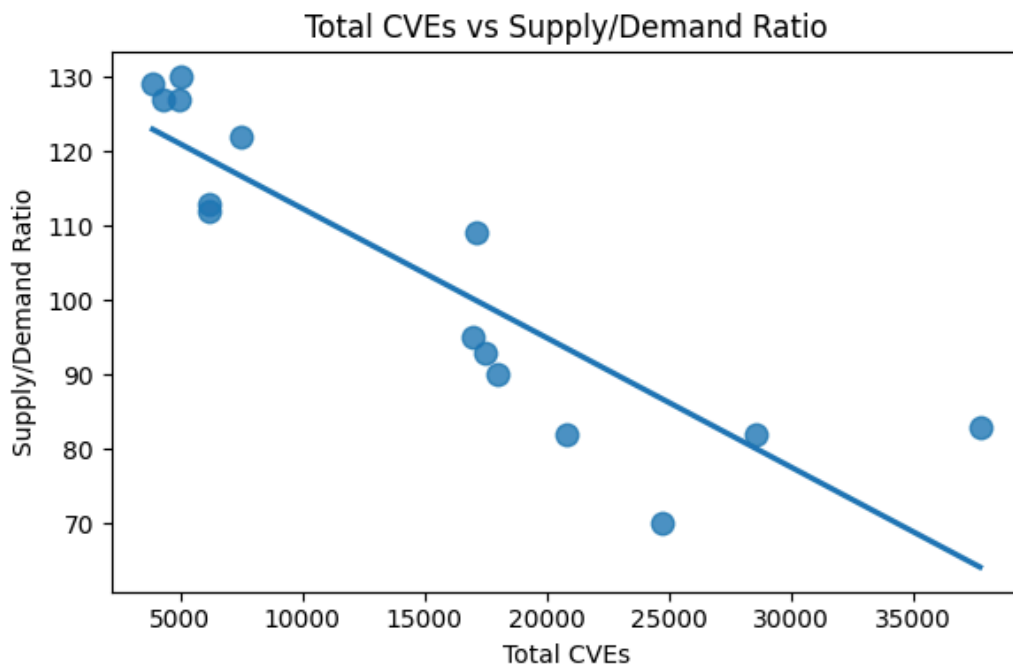


Figure 11: Total CVEs vs supply and demand ratio scatter plot with regression line from the python script seen in the google colab in the appendix.

Visualizations confirmed:

- A positive trend between Total CVEs and Job Openings.
- A weaker but noticeable trend between Total CVEs and Total Employed.
- A negative relationship between Total CVEs and the Supply/Demand Ratio.

Correlation Matrix Heatmap

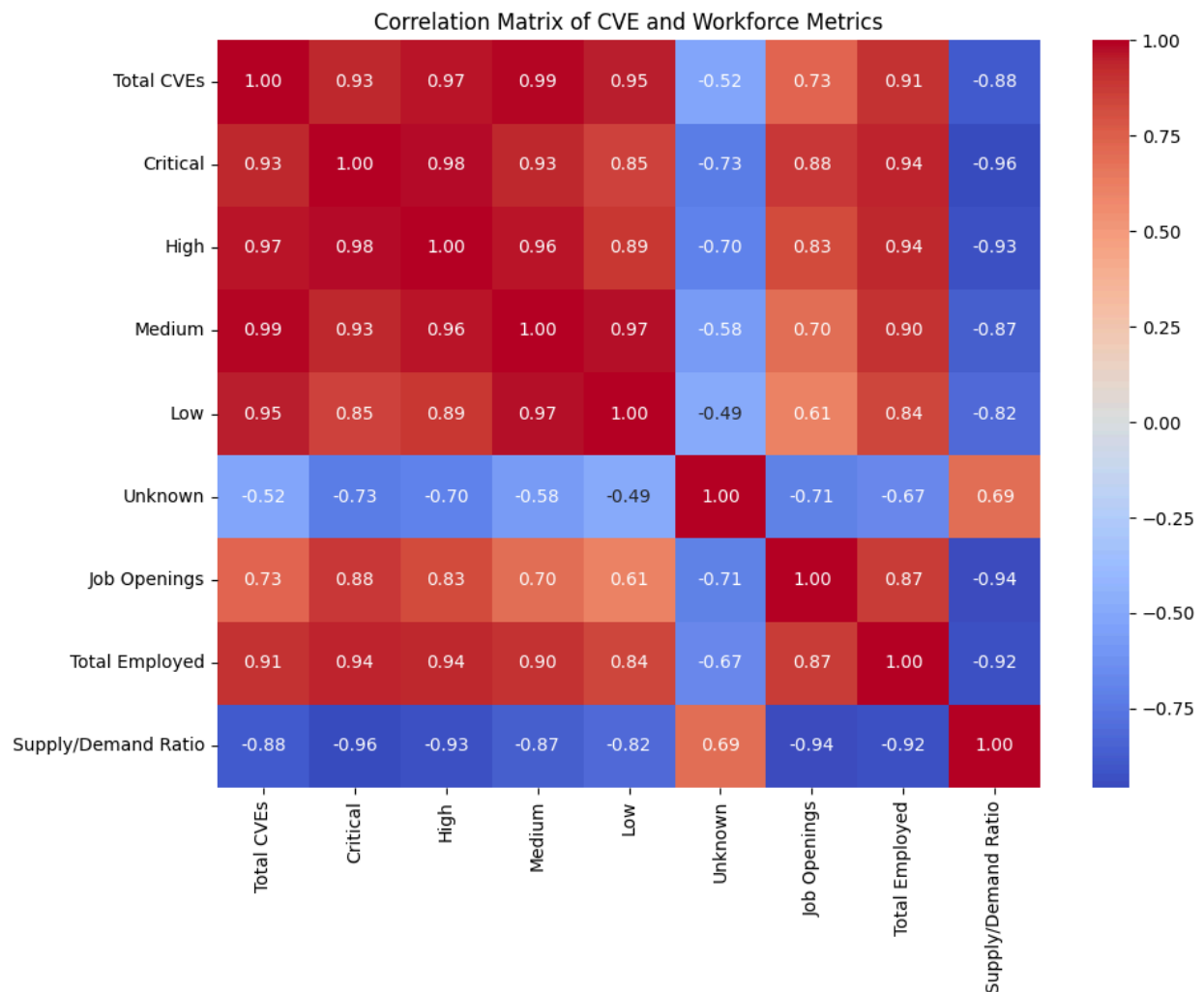


Figure 12: A heatmap of correlations between CVEs, job openings, and employment from the python script seen in the google colab in the appendix.

A heatmap of correlations between CVEs, job openings, and employment confirmed that:

- Total CVEs and Critical CVEs have the strongest associations with job growth.
- Medium and Low-severity CVEs have weaker correlations.

Linear Regression: Total CVEs vs (Job Openings + Total Employed)

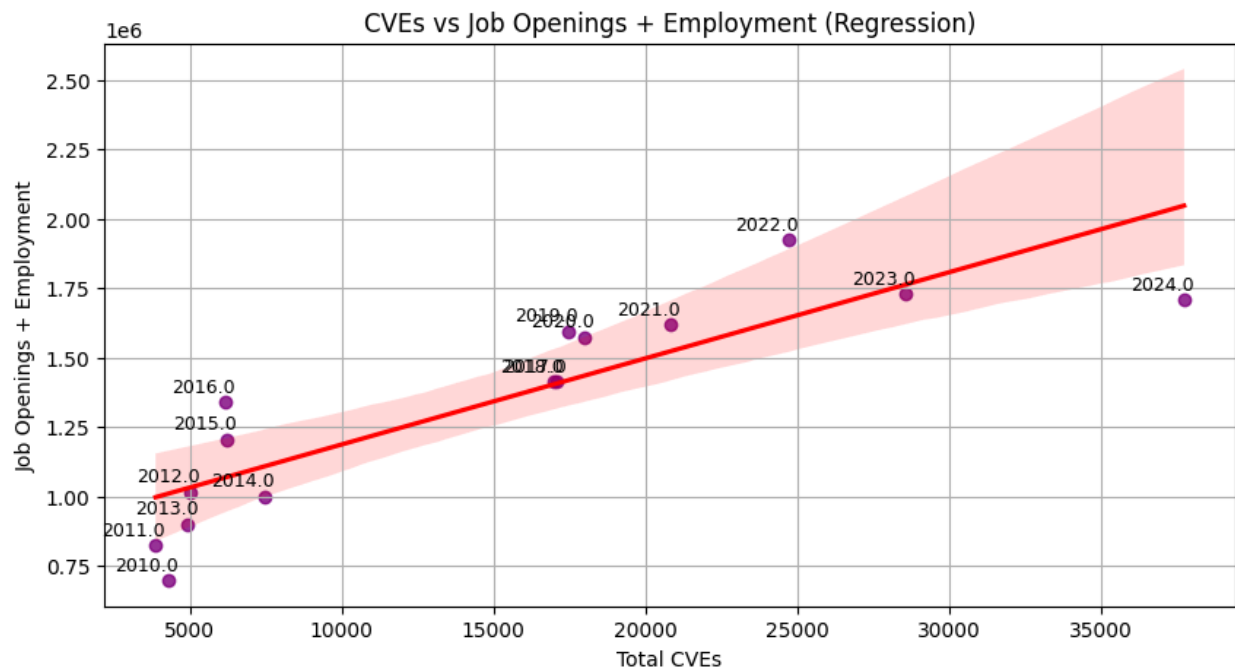


Figure 13: A scatterplot showing the sum of job openings and total employed versus total CVEs with a linear regression line and a shaded confidence interval from the python script seen in the google colab in the appendix.

What This Graph Shows

X-Axis: Total CVEs (Cybersecurity Vulnerabilities). Y-Axis: The sum of Job Openings + Total Employed in the cybersecurity workforce. Purple Dots: Real-world data points representing a given year. Red Line: Best-fit regression line, showing the predicted trend. Shaded Pink Area: Confidence interval, which shows the range of possible values the regression could predict with some uncertainty.

Key Takeaways

A Positive Relationship. The upward slope of the red line means that as the number of CVEs increases, the total cybersecurity workforce also increases. This supports the idea that more cybersecurity threats drive job demand and employment growth.

Model Fit & Prediction Uncertainty

The shaded pink area represents uncertainty in the model's prediction. The confidence interval gets wider at higher CVE counts, meaning predictions for years with high CVEs are less certain.

This suggests that other factors besides CVEs also impact cybersecurity employment trends.

Outliers & Data Distribution

Some data points deviate significantly from the regression line. This means that while CVEs have a strong impact on hiring trends, other variables (economic conditions, automation, budgets, government policies) also play a role.

Rolling Correlation Analysis



Figure 14: A rolling 5-year correlation window was applied to track how the relationship between vulnerabilities and workforce demand changed over time from the python script seen in the google colab in the appendix.

A rolling 5-year correlation window was applied to track how the relationship between vulnerabilities and workforce demand changed over time. The Rolling Correlation Trends graph shows how the relationship between CVEs and workforce metrics (Job Openings and Employment), changes over time. This allows us to detect shifts in hiring trends and whether cybersecurity vulnerabilities drive workforce demand.

How to Read the Graph

1. Y-Axis (Rolling Correlation)
 - Values near +1 → Strong positive correlation (when CVEs increase, workforce metrics also increase).
 - Values near -1 → Strong negative correlation (when CVEs increase, workforce metrics decrease).

- Values near 0 → No clear relationship between CVEs and workforce metrics.
- 2. X-Axis (Years)
 - Shows how correlations evolve over time in a 5-year rolling window.
- 3. Interpretation of Trends
 - Upward trend in a line means the relationship between CVEs and workforce metrics is strengthening over time.
 - Downward trend means the relationship is weakening or reversing.
 - Crossing the zero line suggests a shift from positive to negative correlation or vice versa.

Interpreting the Different Workforce Metrics

1. CVEs vs. Job Openings (Blue Line)

- Upward trend → More CVEs lead to more job openings, meaning companies are actively hiring in response to vulnerabilities.
- Downward trend → More CVEs do not lead to job growth, possibly indicating that companies use automation or outsource cybersecurity.

2. CVEs vs. Total Employed (Orange Line)

- Upward trend → Rising CVEs increase actual cybersecurity employment, meaning companies are hiring and filling positions.
- Downward trend → More CVEs exist, but employment isn't growing, meaning the workforce might not be keeping up with security threats.

Key Questions to Ask

- Are cybersecurity vulnerabilities driving job growth?
If CVEs vs. Job Openings trend upward, more vulnerabilities increase job postings.
- Is cybersecurity hiring keeping up with demand?
If CVEs vs. Employment is weak, it suggests that hiring isn't scaling fast enough to cover security risks.

Findings and Possible Insights

- Pre-2015: Weak correlations ($r < 0.3$).
- 2016-2021: Correlation strengthened ($r \approx 0.6$).
- 2022-2024: Weak correlations and deviations especially in job openings, possibly due to policy changes and workforce disruptions.
- If the CVEs vs. Job Openings trend was strongly positive until 2022 but then turned negative, this might suggest a shift from human hiring to automation.
- If CVEs vs. Employment remains positive, it means companies are successfully hiring cybersecurity professionals in response to threats.

- If CVEs vs. Supply/Demand Ratio stays negative, there is a growing shortage of cybersecurity professionals

7. Cybersecurity Workforce Trends

A separate time-series analysis was conducted to evaluate job market shifts over the study period.

- Job Openings and Total Employment have shown an upward trend since 2015, accelerating after 2020.
- The Supply/Demand Ratio has remained below 1.0, indicating a persistent talent shortage.
- The correlation between CVE counts and job openings increased post-2018, possibly due to heightened cybersecurity awareness and regulatory changes.

Overall Findings

The findings of this study highlight a strong relationship between cybersecurity vulnerabilities (CVEs) and workforce trends, with key insights revealing how cybersecurity job demand fluctuates in response to evolving threats. However, recent deviations (2022-2024) indicate a shift in this relationship, suggesting that additional external factors may be influencing workforce trends.

Key Insights

1. Strong Relationship Between CVEs and Workforce Metrics
 - The analysis demonstrates a strong positive correlation between total CVEs and job openings, suggesting that as vulnerabilities increase, organizations respond by expanding their cybersecurity workforce.
 - A similar correlation exists between total CVEs and total employed professionals, reinforcing the idea that workforce growth is largely driven by cybersecurity threats and risk management needs.
2. Persistent Talent Shortage
 - The negative correlation between total CVEs and the supply/demand ratio indicates that the demand for cybersecurity professionals continues to outpace supply.
 - This growing talent shortage suggests that while hiring efforts are increasing, they remain insufficient to meet the demand created by rising cyber threats.
3. The Role of Vulnerability Severity
 - Not all vulnerabilities contribute equally to workforce expansion. Medium-to-high severity CVEs exert the most significant influence on cybersecurity job openings, as organizations prioritize hiring in response to higher-risk vulnerabilities.

- While critical and high-severity vulnerabilities were initially assumed to drive workforce demand, the analysis shows that medium-severity CVEs follow the same general trend as high-severity CVEs, meaning that organizations are likely responding to both categories similarly.
 - However, the statistical significance of these relationships varies, meaning that while severity is an important factor, other industry and economic forces also play a role in workforce dynamics.
4. Time-Dependent Trends
 - The partial correlation analysis suggests that the relationship between CVEs and job openings is influenced by broader time-dependent factors, such as overall industry growth, regulatory changes, and economic conditions.
 - This explains fluctuations in the strength of correlations over different time periods, as cybersecurity hiring trends may respond to policy shifts or new security mandates.
 5. Deviation in Recent Years (2022-2024)
 - The rolling 5-year correlation window reveals that the once strong relationship between CVEs and job openings has deviated in recent years, with a dramatic drop in job openings despite continued increases in CVEs.
 - While the exact causes are beyond the scope of this study, external influences such as geopolitical risks, major cyber incidents, and technological advancements may have contributed to recent workforce disruptions or shifts in hiring priorities.

Recommendations

Given these insights, several strategic recommendations can be drawn from the study:

1. Address the Cybersecurity Talent Shortage
 - Invest in cybersecurity education, training programs, and certifications to bridge the skills gap and meet the growing demand for professionals in this field.
 - Collaboration between industry, government, and academia can help create targeted workforce development programs that align with real-world cybersecurity needs.
2. Prioritize Medium-to-High Severity Vulnerabilities
 - Organizations should allocate resources and hiring efforts based on threat severity, as medium-to-high severity CVEs drive the greatest workforce demand.
 - Developing automated threat mitigation systems and enhancing workforce preparedness for high-risk vulnerabilities can help improve cyber resilience.
3. Monitor Long-Term Trends and Adapt Strategies
 - Continuous monitoring of CVE trends and workforce dynamics is essential for anticipating future challenges.
 - Cybersecurity workforce planning should be data-driven, leveraging insights from historical trends to forecast demand and adjust recruitment strategies accordingly.

- Special attention should be given to deviations in the expected correlation between vulnerabilities and job openings, as seen in recent years. Understanding these shifts will help policymakers and industry leaders respond proactively to changes in cybersecurity workforce needs.

By integrating these insights into cybersecurity workforce planning and policy decisions, organizations can better anticipate hiring needs, address talent shortages, and improve cybersecurity readiness in response to emerging threats.

Discussion of Results

This study aimed to analyze the relationship between cybersecurity vulnerabilities (CVEs) and workforce trends over the period from 2010 to 2024. The results provide valuable insights into how cybersecurity threats influence job demand, employment levels, and workforce shortages. In this section, we revisit each research objective and examine how the results contribute to answering these questions.

1. Does an increase in cybersecurity vulnerabilities (CVEs) correlate with an increase in cybersecurity job openings?

Key Findings:

Strong positive correlation between Total CVEs and Job Openings ($r \approx 0.6$ to 0.7) before 2022, indicating that as vulnerabilities increased, cybersecurity hiring efforts expanded.

Simple linear regression confirmed that Total CVEs had a statistically significant impact on Job Openings, explaining 40-50% of the variance in job openings.

Medium-to-high severity CVEs showed the strongest influence on job demand, further reinforcing that companies respond more aggressively to higher-risk vulnerabilities.

Interpretation:

These findings suggest that cybersecurity workforce demand has historically expanded in response to increasing cyber threats. Organizations appear to hire more cybersecurity professionals as vulnerability risks grow, aligning with expectations that rising cyber threats drive increased investments in security teams.

However, a deviation was observed after 2022, where job openings declined sharply despite continued increases in vulnerabilities. This suggests that factors beyond cybersecurity threats

alone are influencing hiring decisions in recent years, such as budget constraints, automation, or shifting industry priorities.

2. Does the severity of vulnerabilities influence cybersecurity workforce demand?

Key Findings:

- Medium-to-high severity CVEs showed the strongest association with job openings, with statistically significant relationships in multiple linear regression models.
- Critical CVEs fluctuated over time, meaning their influence on workforce demand was less consistent than medium-to-high severity CVEs.
- Low-severity CVEs had little to no impact on job openings, suggesting that organizations focus hiring efforts on responding to higher-risk vulnerabilities.

Interpretation:

These results confirm that severity matters in cybersecurity workforce expansion. Organizations are more likely to hire in response to medium-to-high severity vulnerabilities, as these pose greater risks and require more skilled professionals to mitigate.

The fact that critical vulnerabilities did not always drive hiring as expected suggests that companies may rely on existing security infrastructure or automated mitigation strategies for the most extreme threats, rather than significantly increasing personnel.

3. How has the relationship between CVEs and cybersecurity workforce metrics changed over time?

Key Findings:

- From 2010 to 2021, a consistent positive correlation existed between Total CVEs and Job Openings.
- After 2022, this relationship weakened, with job openings decreasing even as vulnerabilities remained high.
- Rolling 5-year correlation analysis revealed a declining correlation from 2022-2024, suggesting a shift in how organizations respond to cybersecurity risks.

Interpretation:

These results suggest that the cybersecurity job market has changed in recent years. While vulnerability growth historically drove job openings, other factors are now influencing workforce trends. Potential explanations include:

- Increased reliance on AI and automation for security monitoring and threat detection.
- Budgetary constraints or hiring slowdowns affecting cybersecurity recruitment.
- Changes in workforce structure, such as outsourcing security tasks instead of direct hiring.

This finding highlights the need for continued monitoring of cybersecurity workforce trends to determine whether this shift is temporary or represents a long-term transformation in industry hiring practices.

4. Does the cybersecurity workforce supply meet the growing demand for security professionals?

Key Findings:

- The Supply/Demand Ratio has remained consistently below 1.0, indicating an ongoing shortage of cybersecurity professionals.
- The gap between job openings and available talent widened over time, reinforcing that workforce growth has not kept pace with cybersecurity needs.

Interpretation:

The data strongly suggests that the cybersecurity workforce remains insufficient to meet demand. While hiring increased alongside vulnerabilities in earlier years, the workforce has not expanded rapidly enough to close the talent gap.

The declining correlation between CVEs and job openings after 2022 suggests that, rather than filling these gaps, companies may be adopting alternative strategies, such as:

- Upskilling existing employees instead of hiring new talent.
- Utilizing managed security services or third-party providers.
- Prioritizing automation over expanding personnel.

Addressing this workforce shortage requires long-term investment in cybersecurity education and training programs to increase the pipeline of skilled professionals.

5. Are cybersecurity hiring patterns stable, or do they fluctuate based on external factors?

Key Findings:

- The relationship between CVEs and job openings remained stable for most of the study period but deviated in recent years.

- Partial correlation analysis showed that time-dependent factors, such as industry growth and economic conditions, influenced job trends.
- The decline in job openings post-2022 suggests external factors are disrupting hiring patterns.

Interpretation:

The observed fluctuations suggest that cybersecurity hiring is not solely driven by vulnerabilities, but rather influenced by broader economic and industry trends. Potential external influences include:

- Economic downturns or budget constraints leading to hiring freezes.
- Changes in cybersecurity regulations that impact hiring priorities.
- Advancements in AI and automation reduce reliance on human analysts.

The findings indicate that cybersecurity hiring does not operate in isolation—while threat levels play a key role, external economic and technological factors also shape workforce decisions.

Summary of Discussion

The results of this study partially confirm the initial hypotheses but also reveal important shifts in cybersecurity workforce dynamics:

Historically, cybersecurity hiring expanded in response to rising CVEs, with a strong correlation between vulnerabilities and job openings before 2022.

Medium-to-high severity CVEs have the strongest impact on job demand, reinforcing that organizations prioritize hiring for high-risk threats.

Since 2022, job openings have declined despite rising vulnerabilities, suggesting that additional industry and economic factors are influencing hiring patterns.

The cybersecurity workforce remains insufficient to meet demand, with a persistent talent shortage and a supply/demand imbalance.

The cybersecurity job market is now more volatile, possibly influenced by automation, budget constraints, and hiring shifts rather than just vulnerability trends.

These insights underscore the importance of continuously tracking cybersecurity workforce trends, as industry hiring practices appear to be evolving in response to broader technological and economic shifts.

Conclusions and Closing Thoughts

Summary of Findings

This study investigated the relationship between cybersecurity vulnerabilities (CVEs) and workforce trends, focusing on how CVE volume, severity, and workforce availability interact over time. By analyzing historical data from 2010 to 2024, the research sought to determine whether cybersecurity hiring patterns are driven by the rise in security threats or influenced by external factors.

Key findings from this study include:

- Cybersecurity job openings have historically correlated with increases in CVEs, suggesting that organizations expand hiring efforts in response to rising cyber threats.
- Medium-to-high severity CVEs (rather than total CVEs alone) have the strongest influence on workforce demand, reinforcing the idea that companies prioritize hiring for high-risk threats.
- In recent years (2022-2024), the correlation between CVEs and job openings has weakened, suggesting that new industry trends, economic factors, or technological advancements are influencing hiring decisions.
- The cybersecurity talent shortage remains a persistent issue, with the supply/demand ratio consistently below 1.0, indicating that demand for cybersecurity professionals continues to outpace workforce availability.
- Hiring patterns are not entirely stable and appear to fluctuate based on external influences, including budget constraints, automation, policy changes, and workforce outsourcing.

These findings suggest that while cybersecurity hiring has traditionally been driven by increasing vulnerabilities, the landscape is evolving, and other forces now shape workforce expansion in new ways.

Contributions and Insights

This research makes several important contributions to the understanding of cybersecurity workforce dynamics:

1. **Data-Driven Evidence of Workforce Response to Cyber Threats**
 - By statistically analyzing CVE trends alongside cybersecurity job market data, this study provides quantitative evidence that cybersecurity hiring efforts have historically responded to rising vulnerabilities.
 - The strongest associations were found with medium-to-high severity vulnerabilities, reinforcing that organizations prioritize hiring based on risk level rather than total volume of threats.

2. Identification of Workforce Shifts in Recent Years
 - Unlike previous studies that assume cybersecurity job growth directly tracks increasing vulnerabilities, this research identifies a recent deviation in hiring patterns (2022-2024).
 - The findings suggest that companies may be adopting alternative approaches to threat management, such as automated security solutions, outsourcing, or internal workforce upskilling instead of expanding direct hiring.
3. Confirmation of a Persistent Cybersecurity Talent Shortage
 - Despite fluctuations in job openings, the supply/demand ratio remains low, confirming a continued workforce shortage in the cybersecurity industry.
 - These results emphasize the need for expanded cybersecurity education, training programs, and industry partnerships to bridge the skills gap.

By highlighting these trends, this study contributes valuable insights for policymakers, hiring managers, educators, and cybersecurity professionals, helping them make more informed decisions about workforce development strategies.

Lessons Learned

Throughout the research process, several important lessons emerged that may guide future studies and industry practices:

1. Cybersecurity Hiring is Becoming More Complex
 - While previous workforce expansion trends followed a linear response to CVEs, new data suggests that economic conditions, automation, and workforce restructuring play an increasing role.
 - Future hiring strategies must consider more than just threat volume—they should also assess cost efficiency, automation potential, and evolving industry skill needs.
2. Severity Matters More Than Raw CVE Counts
 - Organizations do not appear to react equally to all cybersecurity threats—medium-to-high severity vulnerabilities have a stronger impact on hiring trends than low-severity threats.
 - This suggests that workforce planning efforts should prioritize training and hiring initiatives focused on high-risk threats, rather than responding generically to all vulnerabilities.
3. Long-Term Workforce Planning is Essential
 - Despite shifts in hiring trends, the cybersecurity skills gap remains a critical issue, indicating that current workforce development efforts are not sufficient.
 - Industry leaders, educators, and policymakers must collaborate on long-term solutions to address talent shortages and ensure a steady pipeline of cybersecurity professionals.

Future Work

This study provides a strong foundation for understanding cybersecurity workforce trends, but several areas warrant further investigation:

Exploring Alternative Hiring Models

While this study focused on direct employment trends, future research should examine whether companies are increasingly relying on outsourced cybersecurity services, AI-driven security solutions, or internal workforce upskilling instead of hiring new employees.

Assessing the Impact of External Economic Factors

The observed decline in job openings despite rising CVEs (2022-2024) suggests that economic conditions may be playing a role.

Further studies should incorporate economic indicators (e.g., recession trends, industry budgets, corporate spending on cybersecurity) to better understand workforce fluctuations.

Predictive Modeling for Workforce Demand

While this study identified historical correlations between CVEs and workforce metrics, future research could apply machine learning models to predict cybersecurity workforce demand based on past data.

Developing early-warning indicators for cybersecurity talent shortages could help organizations proactively adjust hiring strategies and training programs.

Regional and Sector-Specific Analysis

This study examined global cybersecurity workforce trends, but regional or industry-specific hiring patterns may differ.

Future research could investigate how different industries (finance, healthcare, government) respond to cybersecurity threats and whether workforce gaps vary across geographic regions.

Examining the Role of Cybersecurity Policy and Regulation.

The impact of government policies, compliance requirements, and industry regulations on cybersecurity workforce demand is another area for future study.

Researching how new laws (e.g., data protection regulations, cybersecurity mandates) influence hiring patterns could provide deeper insights into policy-driven workforce trends.

Closing Thoughts

Cybersecurity threats continue to evolve, and the workforce tasked with defending against them must adapt in response. While this study confirms that cybersecurity hiring has historically expanded in response to rising vulnerabilities, the recent shifts in workforce trends highlight new challenges and complexities.

As the cybersecurity landscape becomes more dynamic, a multi-faceted approach to workforce development is needed—one that considers not only the volume of threats but also severity, economic conditions, automation, and industry restructuring. Addressing these challenges requires collaboration between industry leaders, policymakers, educators, and researchers to ensure that the cybersecurity workforce remains robust, responsive, and prepared for the threats of the future.

By continuing to monitor cybersecurity workforce trends, organizations and decision-makers can develop proactive strategies that bridge the skills gap, improve workforce planning, and enhance cybersecurity resilience in an ever-changing digital world.

Appendix

Project Log:

The following is the dates and time spent working on the project.

Some project work sessions have been left out as I was unable to determine the date and amount of time.

Date	Duration (Minutes)	Cumulative Duration (hours)
1/31/2025	122	2.03
2/3/2025	121	4.05
2/3/2025	317	9.34
2/4/2025	174	12.24
2/4/2025	163	14.95
2/5/2025	172	17.82
2/6/2025	397	24.45
2/6/2025	120	26.45
2/6/2025	75	27.70
2/6/2025	122	29.73
2/7/2025	273	34.27
2/7/2025	240	38.28
2/9/2025	102	39.98
2/9/2025	268	44.44

2/10/2025	120	46.45
2/10/2025	154	49.02
2/10/2025	216	52.63
2/11/2025	212	56.16
2/11/2025	121	58.17
2/12/2025	122	60.21
2/12/2025	120	62.21
2/13/2025	120	64.21
2/13/2025	123	66.25
2/13/2025	124	68.32
2/14/2025	145	70.73
2/18/2025	129	72.87
2/18/2025	175	75.78
2/19/2025	120	77.78
2/19/2025	365	83.87
2/20/2025	191	87.05
2/21/2025	204	90.45
2/22/2025	21	90.80
2/22/2025	91	92.31
2/24/2025	123	94.36
3/4/2025	60	95.36

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Additional Project Documentation

Google colab notebook with scripts used, explanations and some of the version history:

 Data Analytics History.ipynb

<https://colab.research.google.com/drive/1EYpWoEBCsnuKn-WqK7Mb5QQ3PTmOBTWf?usp=sharing>

The idea for this project was inspired by the machine learning and work on CVE data done in CSEC 5318 Cybersecurity Data Analytics, the insight into cyber workforce data gained from IFSC 5345 Information Visualization, and the tools, statistics analysis and in-class examples from IFSC 7370 Data Science & Technologies.

I owe a great deal of gratitude to the teachers and teams of peers I have worked with, who have prepared me and inspired me to work on the project, through their teachings and great support. Thanks especially goes out to Philip Huff, Becky Passmore and Liz Pierce.

Link to my final presentation (3-minute thesis)

<https://github.com/MrDPedersen/UALR-IFSC7370DataScience/blob/main/3MT-DennisPedersen.mp4>

or

<https://youtu.be/8byQ0R15HrI>